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An Augmented Reality Framework for Visualization of Internet of Things Data for Process Supervision in Factory Shop-Floor

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Abstract

Modern industrial environments provide a high level of digitalization, enabling great volumes of information to be used for shopfloor monitoring and control purposes. In cases where human operators participate in the production process, it is important to support operator's intervention with Internet of Things (IoT) data and empower process supervision with constant information over Key Performance Indicators (KPIs). Towards this end, the proposed framework exploits Augmented Reality (AR) for the provision of an IoT data visualization solution to support KPIs based process supervision in the smart factory environment. The solution utilizes IoT-enabled components, in order to receive dynamic information over the status of sensors and resources spread across the factory shop-floor. The AR application facilitates the need for visualization of information available on the network, on the actual shopfloor components, which offer transparency and ease the supervision of production processes. The development of the AR and cloud API has been based on CSharp and Java, in a reusable framework that can be reused for similar industrial cases. The reconfigurability and reuse of the AR solution is supported by a template configuration, allowing industry practitioners to configure application features and external connections without any additional software development. The solution was validated on a thermoforming process supervision case of a white goods production system. The main goal was to use the AR framework to support operators during the process parameters design phase, and dynamic supervision of the shopfloor environment. It is demonstrated that the proposed system can be used for KPIs monitoring of the production cycles and for empowering process supervision and optimization.

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Keywords: Augmented Reality; Data Visualization; Process Supervision; Internet of Things;Digitalization of industry

1. Introduction

Since Industry 4.0 took over in the manufacturing environments, increasingly more production assets are enriched with a digital identity within the so-called Internet of Things (IoT). Engineers have managed to exploit this network of information in order to monitor and optimize production operations and ensure high quality products [1]. Nevertheless, human workforce is still a vital asset in the manufacturing procedure, and the smooth interference between the cyber physical system and human operator is crucial for a seamless production function. To this end, organizations engage actively into research activities that use innovative solutions in order to increase information transparency across the shopfloor environment [2]. Augmented Reality (AR) is a novel human–computer interaction tool that overlays computer-generated information on the real world environment [3] and has been found useful in multiple industrial applications.

AR applications have been involved into vocational training, monitoring, maintenance support, layout management, online guidance and many more manufacturing applications [3], [4]. AR technology has established its way through the classroom, emerging the Teaching Factory paradigm, with knowledge exchange between academia and industrial stakeholders. The ability of borderless communication using advanced visualization technologies has

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given both industry and academia a collaborative framework for training and troubleshooting in real industrial problems [5]. Similarly, collaborative product development is now empowered with AR holographic content, enabling engineers with more realistic representations for remote analysis and design [6] give a highly realistic representation of a space configuration, with digitally projected geometrical models overlaying in real-scale sizes to enable the engineer foresee objects' arrangement in the production before moving on with the actual deployment [7].

Operator support receives the highest interest in modern AR applications in manufacturing, while multiple research studies have been conducted in order to achieve worker-centered information exchange and user satisfaction [8]. The merging of AR technologies in both software and hardware, empower developers to encapsulate a highly realistic and comfortable holographic content to the user's field of view, and provide guidance to multiple operations [9]. The augmented environment has also a beneficial impact on human-robot collaboration processes within hybrid with the projection of process-related information and safety instructions in real-time [10]. On top of that, in maintenance operations, real-time connection between an expert engineer and the shopfloor operator can be established through the network, enabling the holographic content to be dynamic [11], which in some cases promises reduction in the costs of maintenance and Mean Time to Repair of the machine.

Moreover in the past two years several AR applications have been developed to provide a link between IoT information and shopfloor operators, which supports monitoring of production processes as well as online decision-making [12]. The industrial benefits of these methods are high, as within the concept of smart factory results from process simulations and raw sensor information can be broadcasted directly to the human operator [13]. The operator is then more capable of understanding the condition indicators of the system and in cases where digital controllers are involved to provide a feedback loop directly from the AR interface [14]. Such industrial solutions create a source of information exchange across operators and IoT components, thus providing shopfloor information transparency. Although these methods are well known for increasing production efficiency along with many more manufacturing attributes, there are mainly applied in research activities and are not frequently distributed as product solutions on the market [15]. Nevertheless, the high demand in manufacturing industries, for easier shopfloor supervision and early-stage process parameters' design, have increased the interest of these applications in the research community [16].

However, the demands for increased granularity/ reconfigurability and comfort in the user's interfaces, remain the main obstacles into the provision of a widely acceptable and sustainable solution. The proposed research paper is focusing on addressing these challenges with the provision of an easily reconfigurable AR framework for shopfloor supervision, that aims at reducing the use-case deployment cost and duration while at the same time increasing the comfort in the way that operators are interacting with production processes. To do so, the solution is consisted of three main parts: the configuration methodology, the process information streaming mechanisms, and the variety of visualization components available for representing information of different kinds. These aspects provide granularity to the framework, thus enabling the fast and cost-free adaptation of the system in different industrial cases, and advanced visualization and interaction interfaces. The visualization components were developed to display information of multiple types, while the backend-server API was then responsible for storing, handling, and analyzing information on the background, which were accessible at all times by the AR application. The backend information services as well as the advanced visualization interfaces were proven to be very consisted in the stability and comfort of the AR application.

2. Process Supervision Framework

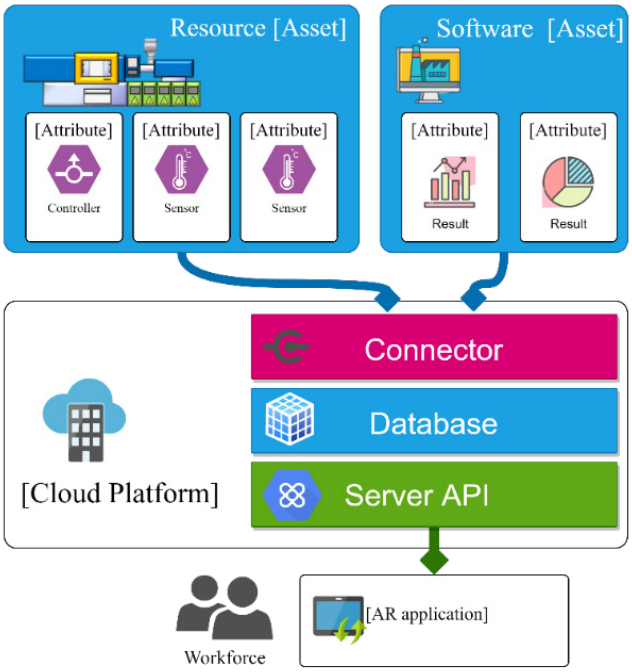

Figure 1. Process supervision framework pipeline

The proposed framework (shown in Figure 1) presents the IoT-enabled production assets, the backend server, and the shopfloor AR application for the human operator. IoT components are modeled as assets which may contain one or more attributes, and can be of various types such as machines, field devices or even process analysis software and services. Information are provided from IoT-enabled components through the cloud server and retrieved by the AR application in order to be displayed to the operator. The server system is responsible for storing information while also providing fast management and retrieval capabilities through the developed APIs. This provides the AR application a way of reducing the storage requirements while also giving to the user the ability to retain any historical information that are stored within the server's database. The application then uses various types of visualization components in order to project information coming from production assets directly to the user interface which will be the main source of information for supervising the production.

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2.1. AR Visualization Components

The AR application utilized several visualization components for displaying IoT information. The following methods of displaying information (Figure 2) have been developed within the application core modules and were instantiated as AR objects directly within the operator's field of view. The proposed application objects have been developed with C# programming language (.NET Framework) and Unity 3D Engine.

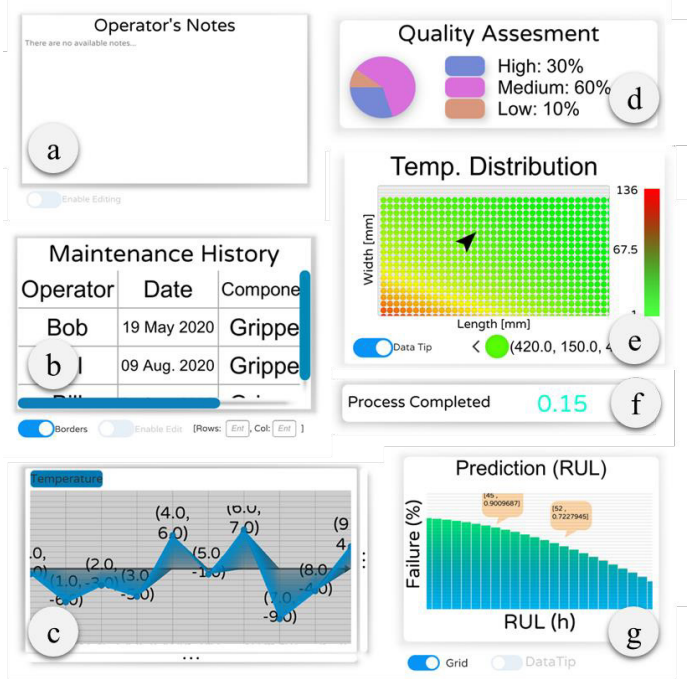

Figure 2. Visualization components: (a) document, (b) datasheet, (c) 2D plot, (d) pie chart, (e) 2D contour, (f) single value, (g) histogram

The *document* component is able to display any kind of textual or image information. Similar to any text editing tool, the user could interact with this content and update the text back to the server. This was mainly used for leaving notes directly on the different production assets to be available by next production operators. The *pie chart* component is used for displaying production attributes characterized by percentage values. The *contour* on the other hand is used for displaying three dimensional vectors which could be for example the distribution of a specific value over a plane. This module is useful in industries working with sheet metal/ plastic raw material, as the output thickness, temperature, pressure etc. could reveal the quality of a given product. Furthermore, *datasheet* is used in order to provide information to the operator in a tabular mode. This module can interact with the user, meaning that the operator could update the content stored on the server directly through the AR application. Main application of this display type is to describe different properties of a system, as well as to document any maintenance operation. *Single value* component can be used for presenting the label of an attribute as well as the value of the item. *3D plot*, is used to display time-related values within a timeseries graph. The user is also able to navigate through the previous values, select the number of historical instances that wants to be stored on the graph and inspect some statistical values (mean, max, min, standard deviation) of the data as a way of revealing the current trend of the attribute. Finally, the *histogram* is used in order to display the frequency distribution of an attributes value based on a statistical analysis.

Updating the content of each visualization component requires information to be provided within the right format supported by the component's scripting interface. Based on the visualization type the input format was different as shown in Table 1. Missing to provide information within this format would return an error exception to the internal application handlers. As a way of minimizing this limitation, flexible data handling mechanisms have been developed, enabling the system to identify the required input format even in cases it is not provided exactly as expected.

Table 1. Input format for the different types of displays

Display	Input Format
Document	{title: "string", content: "string"}
2D Plot	{datetime: timestamp, value: float}
2D Contour	List[{x: float, y: float, z:float}]
Pie chart	List[{label: string, value: float}]
Histogram	List[{label: string, value: float}]
Datasheet	List[{col: int, row: int, value: string}]
Single value	{label: string, value: string}

2.2. Application Interfaces

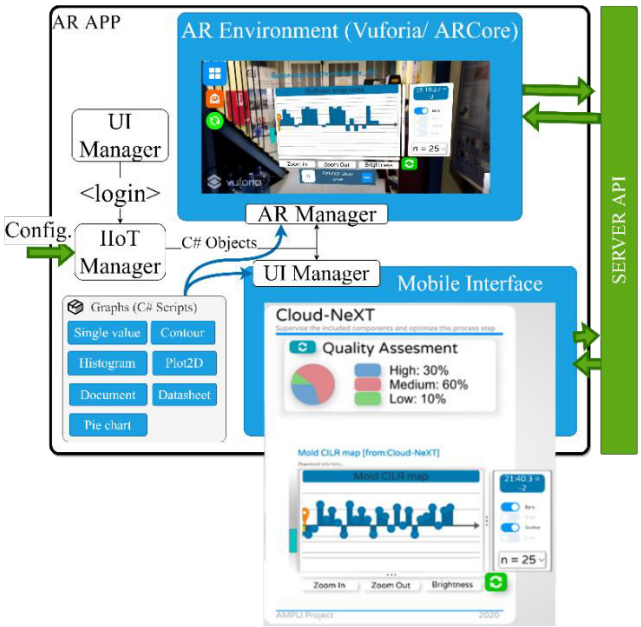

Figure 3. Internal application architecture

The AR application infrastructure is empowered with two different modes of operating: the AR supervision mode and on-screen monitoring interface as shown in Figure 3. The proposed interfaces design derived from case-specific requirements analysis and continues fine-tuning development updates during the prototypical implementation of the proposed application. The AR supervision mode enables the user to monitor IoT data

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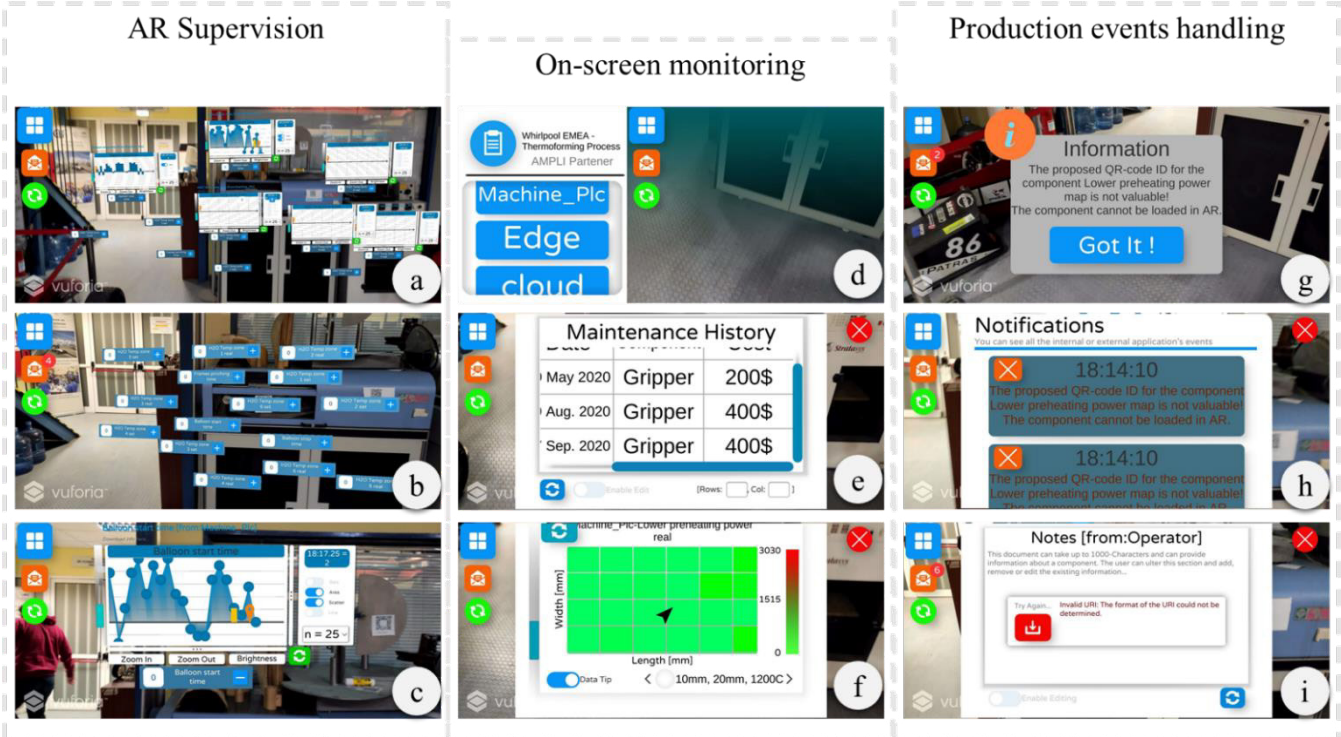

Figure 4. Screenshots from the AR application interfaces

by projecting the different visualization components over the real-world objects on the shopfloor. Based on the scenario definition, the "*AR manager*" module instantiates the environment using the available components of the previous section. Vuforia and ARCore were used within Unity 3D Engine to enable spatial understanding and image recognition for projecting the AR components over the physical space. Operators are able to view information over the actual production assets, directly by moving the device over the asset's QR-code (Figure 4: a, b, c).

The second mode (shown in Figure 4: d, e, f) provides the same information content without the need for constructing the AR environment. This is needed in order to provide quick access to all the available IoT data without the need for moving to the location of the industrial equipment. The operator is thus able to open a side menu containing all the available assets on the factory and quickly navigate in all the available attributes from a scroll-down page on the tablet's screen.

Finally the application offers events and error handling data (Figure 4: g, h, i) that are displayed to the operator in order to monitor production events and manage external errors or incompatibilities. Notifications are displayed over the visualization components, as well as on the notifications panel. Notifications could be in type of production events (e.g. process temperature is out of the expected limits) or software errors (e.g. lost connection to the server). Information in type of production events is also displayed within a pop-up notification window to the user's screen, in order to enable operator reaction in a timely manner over the issue.

2.3. Configuration & formatting

The application has been developed to be reconfigurable for different production cases directly from a configuration file. This file describes all the external connections where the application can retrieve information and provides a detailed description over the assets and attributes of production case. Operator provides the credential linked to the specific production while login in and the AR application retrieves the configuration file for the specific production scenario by the server's API. The configuration file follows the bellow template represented within a JSON file format:

```
{
  "name": "Production Name",
  "connector": {
    "endpoint": "http://data-connector-api",
    "past_duration": 5,
    "commands": [ "load_plc_data", "load_simulat" ]
  },
  "IoT_components": [
    {
      "id": "01",
      "asset": "Machine_PLC",
      "attribute": "Motor drivers temperature",
      "QR_code": "QR Thermoforming Machine",
      "download_frequency": 20,
      "end-point": "http://api/asset/attribute",
      "topic": "topic-name",
      "protocol": "(HTTP) GET-POST/(BROKER) PUB-SUB",
      "data_pointer": "node1:node2:value",
      "display_type": "Plot2D",
      "display_location": {
        "x": 0,
        "y": 0,
        "z": 0
      }
      "value_limits": {
        "upper": 100,
        "lower": 20
      }

    }
  ]
}
```

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The "*connector*" field of the file creates a link between the AR application and a cloud-based connector that passes information from local IoT assets to the server's database. A list of functions/ commands for the cloud-connector were provided by the "*commands*" list in order to enable the configuration of multiple services. Each time the user presses the green "refresh" button these commands would be passed to the connector in order to upload information to the backend server in real-time. The "*past_duration*" element defines the historical interval of previous values that the connector would try to handover to the platform.

The "*IoT_components*" list provides all the information of the assets/ attributes for a specific use-case scenario. The "*QR-code*" defines in which of the available QR-codes this specific object is included in. This way, the supervision process of different production resources could be distributed into different QR code images across the shopfloor. Furthermore, the "*download_frequency*" was responsible for defining the sample rate on which the application would request information from the server's API. This functionality was only enabled in cases where the "*protocol*" field was set to HTTP GET-POST. In cases that a broker interface was provided as a message-based middleware for information exchange, the application receives information by subscribing to a specific topic. Finally the "*endpoint*" provided the public IP, for the discovery of the server API.

Furthermore, the "*display_type*" defines the type of the AR visualization component that is going to be used for the specific attribute. One of the visualization components presented in section 2.1 can be provided in this field. To increase the granularity of the proposed solution, the "*data_pointer*" is included. This field is responsible for dynamically discovering the location of the required input information within any arbitrary JSON file model. This allowed the application to work with different type of inputs given from external APIs. The methodology that was used for these purposes is based on discovering the information path by stating the JsonNode of "*parent*" and "*child*" by using the character ":". Similarly in cases where lists are involved, the term "*[list]*" was provided, and the statement for the amount of entities that are going to be checked. Figure 5 illustrates the parsing mechanism from a given input file provided by the server API.

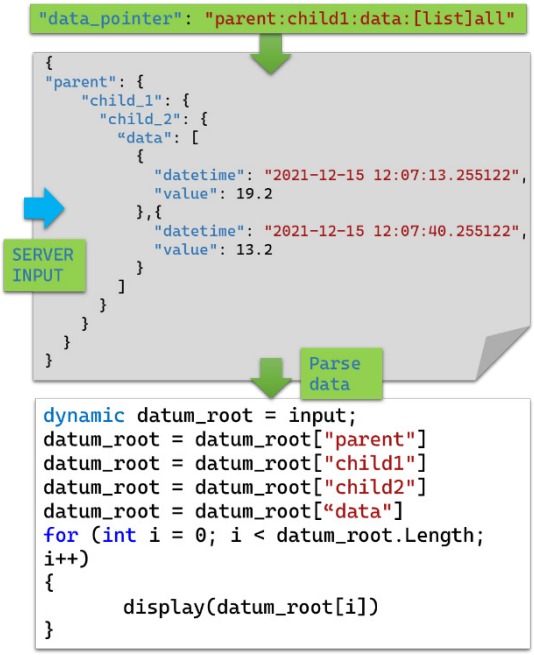

Figure 5. Dynamic parsing of input data based on configuration

3. Pilot case validation

The application has been validated in a real industrial case of a white goods production system. The production requirements indicated the use of the AR supervision framework as a tool for defining the optimal process parameters to a new process as well as use of the AR application after the installation of the new process in order to supervise the efficiency of the system. The process is based on a thermoforming machine, where multiple plastic sheets are entering the machine, heated, and by using vacuum pressure are deformed over the product's mold. The production operators should monitor up to 180 different attributes coming from multiple assets i.e. machine PLC, simulation software, thermal camera and more, while also optimize the process parameters based on the product's quality. To do so, the operators' needs for comparing the actual process results for a given set of input parameters as well as multiple simulation and quality assessment software. This process is a difficult

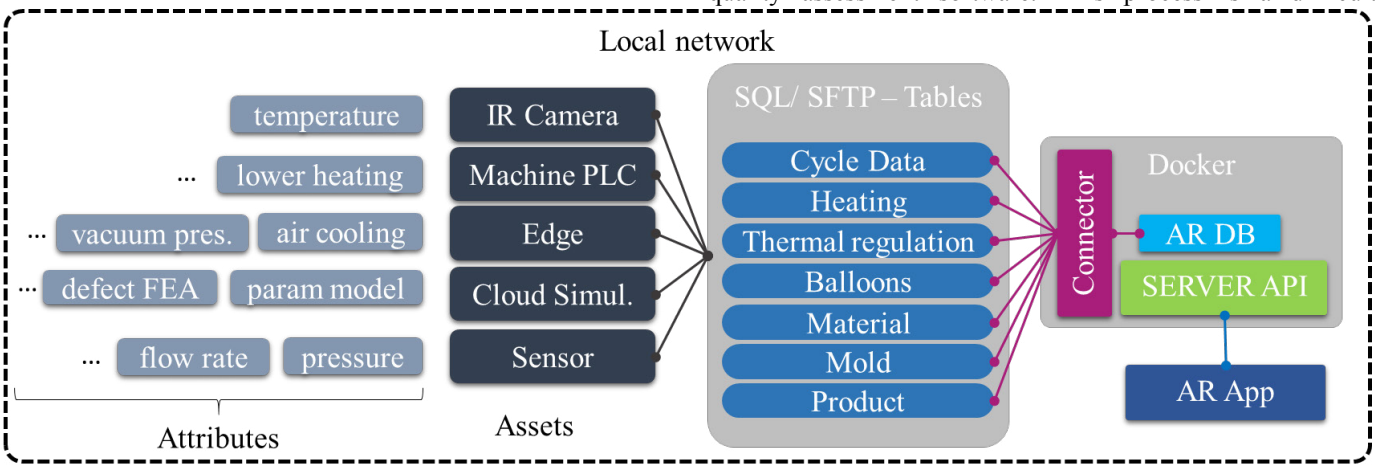

Figure 6. Deployment of AR supervision framework within the use-case architecture

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procedure for production operators especially during the launch of a new product/ production environment.

In the specific production case, information from the thermal camera was uploaded in a SFTP server, while all information coming from the other assets were fed into an SQL database. A cloud-based connector was developed in order to upload information from the SQL/SFTP servers onto the AR backend platform. The reason for putting this additional step, instead of letting the AR application retrieve information from the SFTP/SQL servers was due to the fact that the current version of the AR application is not empowered with any queries mechanisms and relay on an existing API for retrieving information. The components included in the use case scenario are illustrated in Figure 6.

The framework was utilized in day-to-day supervision processes during the mass production launch of a new product. The validation period lasted a week in which three different production operators utilized the AR supervision application. The results showed an appreciated impact on reducing the initial process parameterization duration from approximately two-weeks into three days while also helped in supervising the production performance after the initial launch. Feedback from production operators' questionnaires revealed a good level of satisfaction into the provided user interfaces while also managed to achieve a stable AR environment, avoiding operator fatigue due to uncomfortable interfaces.

4. Conclusion

The proposed research work focuses on the development of an AR framework that could be used in order to support shopfloor operators in process supervision. The framework focuses on creating transparency across shopfloor production processes and IoT components, enabling the operator to perform more efficiently. The proposed AR application along with the information services were developed in such a way that the management of different production aspects in a comfortable and user-friendly experience. The visualization functionalities have been developed based on basic data rendering mechanisms, while for the information exchange HTTP GET/POST and broker Pub-Sub protocols were established. The reconfigurable attribute of the proposed solution was also a high priority during the development, allowing the deployment across different applications to be development-free and fast in order to create a sustainable tool for a wide range of industrial cases.

The proposed solution however introduces several limitations regarding the system's intelligence and visualization methodologies. In specific, additional methods for displaying information might be required, such as 3D graphs, Finite Element Analysis (FEM) analysis results over the actual product and many more based on the use case needs. Future research studies could also enable the system to learn from previous information provided by the user and propose possible actions in different production situations.

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